

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1708

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I N. Joe Jeevish / 19252437 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

N. Joe Jeevish

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20

B

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

1). Smart Home Automation System

It is an Artificial Intelligence application that automatically controls lighting temperature and security by observing the environment and user behaviour. It receives inputs through sensors such as temperature sensor, motion detector, etc and processes the information and decide the appropriate actions and perform the actions through actuators. Since to adopt the environment and respond immediately Intelligent agent such as Simple reflex, model based, goal based and utility based agents

Simple Reflex:

It is a Intelligent agent which selects action based on the current input without considering previous events or future consequences. It follows predefined if-then rules.

Ex:

If motion is detected in the room then turn on the Switch light

If motion is not detected for 10 minutes or more then turn off the light

⇒ It gives fast response but cannot remember previous actions

Model Based Reflex Agent:

It maintains an internal memory of the environment.
It remembers both the current sensor input and previous information to make better decision

Ex:

Remembers the room is occupied recently. It also remembers temporary absence does not mean the room is vacant

It handles partially observable environment but need additional memory

Goal Based Agent:

It chooses actions that can help achieve a specific objective or goal

Ex: Maintain comfortable room temperature. Reduces the electricity consumption.

Utility Based Agent:

Selects the action that provides the highest overall benefit when multiple goals conflict

Ex: Suppose the user wants room to be cooled 24°C it plans accordingly to reduce the electricity bill and run

Best agent for Smart home is hybrid approach Combine Model based, Goal based, utility based agents which is more effective. Model based remembers previous actions, Goal based plans to achieve a specific goal and utility based is used to achieve the best benefit among various possibilities. So a hybrid approach combining these agent would be the best.

2). Robot Navigation in an unknown Maze.

The robot which is placed in an unknown maze needs to find the best path to exit having the prior knowledge of the maze route. Uniform cost search algorithm and iterative Deepening ~~and~~ Search algorithm can be used to find the exit.

UDS

It is used expands the node with lowest cumulative path cost from the starting position. It guarantees the least cost path if all the step cost are positive.

Working

⇒ The robot starts from entrance

- ⇒ It records the cost of ~~see~~ by reaching every path
- ⇒ Among all possible path it finds the lowest cost
- ⇒ The process continue until it reaches exit point

It complete if path cost is positive but requires large memory

IDS:

It repeatedly performs DFS with increasing depth limits until the goal is found

Working:

- ⇒ The robot explores
Depth 0, Then depth 1, depth 2, depth 3
- ⇒ The process continue until it reaches the exit
- ⇒ The robot gradually explores deeper level without storing every explored node.

It ~~is~~ use very low memory but does not guarantee the least cost path when step cost differ

Relation to Intelligent Agent.

The robot acts as an Intelligent agent perceiving the environment through Sensors, process the information, decide the best action, acts using actuators. The

Simple Reflex agent ✓